P1 Let us write H so that the sign of J_1 is apparent:

$$H = \sum_n \left(-J_1 \vec{S}_n \vec{S}_{n+1} + J_2 \vec{S}_n \vec{S}_{n+2} \right) \quad \text{, with } \underline{J_1, J_2 > 0}.$$

$$\vec{S}_n = \frac{1}{\sqrt{N}} \sum_q \vec{S}_q \, e^{i\vec{q}\vec{R}_n}$$

$$\Rightarrow H = \sum_q \underbrace{\left(\frac{1}{2} \sum_{r=\pm1,\pm2} J(r) e^{iqr} \right)} \vec{S}_q \cdot \vec{S}_{-q}$$

$$J(q) = -J_1 \cos(qa) + J_2 \cos(2qa)$$

Then $E = NS^2 J(Q)$ where Q minimizes $J(q)$:

$$\frac{\partial J(q)}{\partial q} = 0 = a\left(J_1 \sin(qa) - 2J_2 \sin(2qa) \right)$$

$$\Rightarrow \sin(qa)\left[J_1 - 4J_2 \cos(qa) \right] = 0$$

$$\sin(qa) = 0 \quad \text{or} \quad \cos(qa) = \frac{J_1}{4J_2} = \frac{1}{4W}$$

$$\cancel{\frac{\partial^2 J(q)}{\partial q^2} = a^2 \left(J_1 \cos(qa) - 4J_2 \cos(2qa) \right)}$$

Note that $\cos(2x) = 2\cos^2 x - 1$

$$\Rightarrow J(q) = -J_1 \cos qa + 2J_2 \cos^2 qa - J_2 =$$

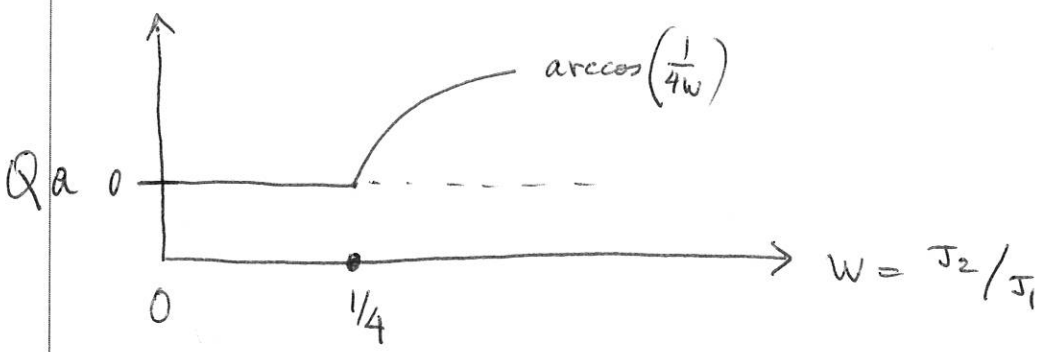
$$= 2J_2 \left(\cos^2 qa - \frac{J_1}{2J_2} \cos qa \right) - J_2 =$$

$$= 2J_2 \left[\left(\cos qa - \frac{J_1}{4J_2} \right)^2 - \left(\frac{J_1}{4J_2} \right)^2 \right] - J_2 =$$

$$= 2J_2 \left(\cos(qa) - \frac{J_1}{4J_2} \right)^2 - J_2 - \frac{J_1^2}{8J_2}$$

Clearly the minimum is at $\cos(qa) = \frac{J_1}{4J_2}$ when possible $\Rightarrow$ that is for $J_2 > \frac{1}{4} J_1$.

For $J_2 < \frac{1}{4} J_1$ the minimum is at $q = 0$.

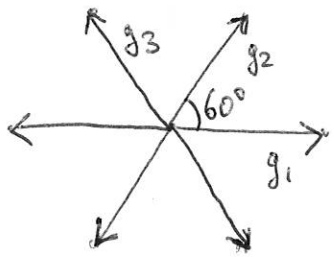


$$Qa = \begin{cases} 0 \ , & \frac{J_2}{J_1} < \frac{1}{4} \\ \arccos\left(\frac{J_1}{4J_2}\right) = \arccos\left(\frac{1}{4w}\right) \ , & \frac{J_2}{J_1} > \frac{1}{4} \end{cases}$$

$$\frac{E}{NS^2} = \begin{cases} -J_1 + J_2 \ , & \frac{J_2}{J_1} < \frac{1}{4} \\ -J_2 - \frac{J_1^2}{8J_2} \ , & \frac{J_2}{J_1} > \frac{1}{4} \end{cases}$$

P2 Triangular lattice antiferromagnet

$$H = \frac{1}{2} J \sum_{r,g} \vec{S}_r \cdot \vec{S}_{r+g}$$



$$\vec{g} = \{ \pm\vec{g}_1, \pm\vec{g}_2, \pm\vec{g}_3 \}$$

nearest neighbor vectors
$$\begin{cases} \vec{g}_1 = (1, 0)\, a \\ \vec{g}_2 = \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right) a \\ \vec{g}_3 = \left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right) a \end{cases}$$

$E_0 = NS^2 J(Q)$ where $\vec{Q}$ minimizes

$$J(q) = \frac{1}{2} \sum_{\pm g_1, g_2, g_3} J\, e^{i\vec{q}\cdot\vec{g}_i} = J\Big(\cos q_x a + \underbrace{+ \cos(\vec{q}\cdot\vec{g}_2) + \cos(\vec{q}\cdot\vec{g}_3)}_{\cos\left(\frac{q_x}{2} + \frac{\sqrt{3}}{2} q_y\right)a + \cos\left(\frac{q_x}{2} - \frac{\sqrt{3}}{2} q_y\right)a = 2\cos\left(\frac{q_x a}{2}\right)\cos\left(\frac{\sqrt{3} q_y a}{2}\right)}\Big)$$

Denote $q_x a \equiv x$, $q_y a \equiv y$ for brevity:

$$J(\vec{q}) = J\left(\cos x + 2\cos\frac{x}{2}\cos\frac{\sqrt{3}y}{2}\right)$$

Find minimum:
$$\frac{\partial J(q)}{\partial x} = 0 = -\sin x - \sin\frac{x}{2}\cos\frac{\sqrt{3}y}{2} \quad (1)$$

$$\frac{\partial J(q)}{\partial y} = 0 = -\sqrt{3}\cos\frac{x}{2}\sin\frac{\sqrt{3}y}{2} \quad (2)$$

From (1): $\sin\frac{x}{2}\left(2\cos\frac{x}{2} + \cos\frac{\sqrt{3}}{2}y\right) = 0 \quad (1')$

$\cos\frac{x}{2} = 0$ is not minimum (would give $J(q) = -J$ only)

$\Rightarrow \sin\frac{\sqrt{3}y}{2} = 0 \Rightarrow \frac{\sqrt{3}}{2}y = \pi n \Rightarrow y = \frac{2\pi n}{\sqrt{3}}$

$\Rightarrow$ from (1'): $2\cos\frac{x}{2} = -\cos(\pi n)$

$$\cos\frac{x}{2} = -\frac{1}{2}\cos(\pi n).$$

Then $\frac{J(\vec{q})}{J} = 2\cos^2\frac{x}{2} - 1 + 2\cos\frac{x}{2}\cos\frac{\sqrt{3}}{2}y =$

$$= 2\cdot\frac{1}{4} - 1 - \cos(\pi n)\cos(\pi n) =$$

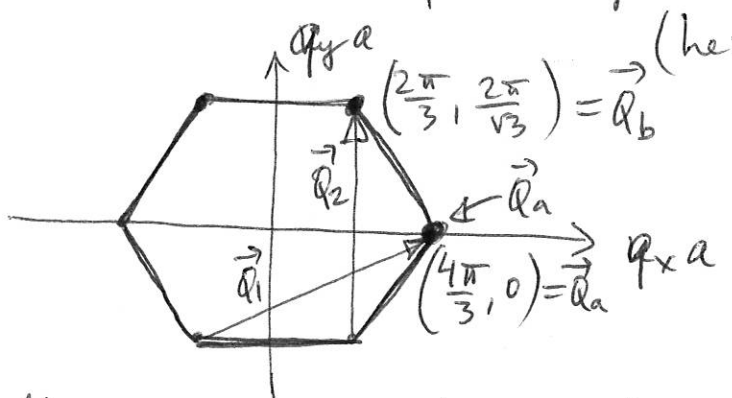
$$= \frac{1}{2} - 1 - 1 = -\frac{3}{2}.$$

Thus $\underline{n=0}$: $\cos\frac{x}{2} = -\frac{1}{2} \Rightarrow \frac{x}{2} = \frac{2\pi}{3} \Rightarrow \underline{x = \frac{4\pi}{3},\ y = 0}$

$\underline{n=1}$: $\cos\frac{x}{2} = \frac{1}{2} \Rightarrow \frac{x}{2} = \frac{\pi}{3} \Rightarrow \underline{x = \frac{2\pi}{3},\ y = \frac{2\pi}{\sqrt{3}}}$

(All other n's give the same two solutions.)

Next work out BZ of triangular lattice: (hexagon!)



All other vertices of hexagonal BZ can be obtained by reciprocal lattice translations: $\vec{Q}_1 = \left(2\pi, \frac{2\pi}{\sqrt{3}}\right)$

$\vec{Q}_2 = \left(0, \frac{4\pi}{\sqrt{3}}\right)$

Check that $J(\vec{q}) = J(\vec{q} + \vec{Q}_{1,2})$

Thus $\begin{cases} \vec{Q}_a = \left(\frac{4\pi}{3a}, 0\right) \\ \vec{Q}_b = \left(\frac{2\pi}{3a}, \frac{2\pi}{\sqrt{3}a}\right) \end{cases}$

$$E_0 = -\frac{3}{2} N S^2.$$

<u>let</u> $\vec{Q} = \vec{Q}_a = \left(\frac{4\pi}{3a}, 0\right)$

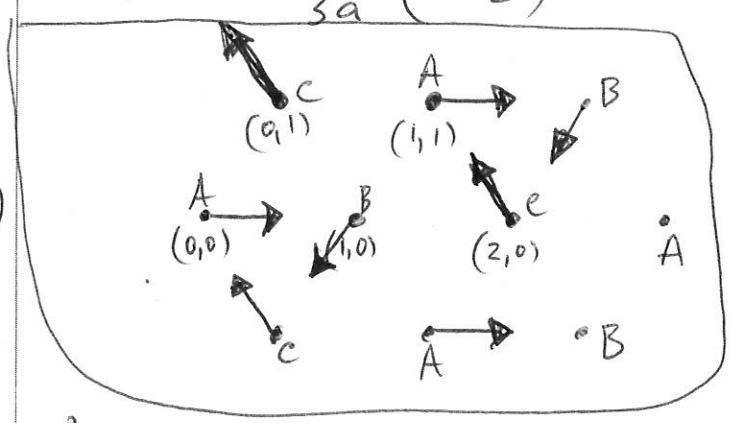
Spin structure $\vec{S}_R = S\left[\vec{n}_1 \cos(\vec{Q}\cdot\vec{R}) + \vec{n}_2 \sin(\vec{Q}\cdot\vec{R})\right]$

where we can choose $\vec{n}_1 = \hat{x}$, $\vec{n}_2 = \hat{y}$.

$$\vec{R} = n\vec{g}_1 + m\vec{g}_2 = \left(n + \frac{m}{2}\right)a\vec{e}_x + \frac{\sqrt{3}}{2} ma\vec{e}_y.$$

$$\vec{Q}\cdot\vec{R} = \frac{4\pi}{3a}\left(n + \frac{m}{2}\right)a + 0\cdot\frac{\sqrt{3}}{2}m = \frac{4\pi}{3}\left(n + \frac{m}{2}\right)$$

(n, m)



There are only 3 orientations of spin: can be described as a 3-sublattice structure (A, B, and C).

[The other choice, $\vec{Q}_b$, gives the same structure upto global rotation.]

$$\frac{\vec{S}}{S}(R=0,0) = \vec{n}_1 = \hat{x}$$

$$\frac{\vec{S}}{S}(R=1,0) = \vec{n}_1 \cos\frac{4\pi}{3} + \vec{n}_2 \sin\frac{4\pi}{3} = -\frac{1}{2}\vec{n}_1 - \frac{\sqrt{3}}{2}\vec{n}_2$$

$$\frac{\vec{S}}{S}(R=0,1) = \vec{n}_1 \cos\frac{2\pi}{3} + \vec{n}_2 \sin\frac{2\pi}{3} = -\frac{1}{2}\vec{n}_1 + \frac{\sqrt{3}}{2}\vec{n}_2$$

$$\frac{\vec{S}}{S}(R=1,1) = \vec{n}_1 \cos 2\pi + \vec{n}_2 \sin 2\pi = \vec{n}_1$$

$$\frac{\vec{S}}{S}(R=2,0) = \vec{n}_1 \cos\frac{8\pi}{3} + \vec{n}_2 \sin\frac{8\pi}{3} = -\frac{1}{2}\vec{n}_1 + \frac{\sqrt{3}}{2}\vec{n}_2$$

$\boxed{P3}$

$$\text{HP bosons}\begin{cases} S^- = \sqrt{2S}\,\hat{a}^+\sqrt{1-\frac{\hat{n}}{2S}} \\ S^+ = \sqrt{2S}\sqrt{1-\frac{\hat{n}}{2S}}\,\hat{a} \\ S^z = S - \hat{n} \end{cases}$$

where $\hat{n} = \hat{a}^+\hat{a}$

$[\hat{a}, \hat{a}^+] = 1 = \hat{a}\hat{a}^+ - \hat{a}^+\hat{a}$

Check: $[S^+, S^-] = 2S^z$

$$(2S)\left[\sqrt{1-\frac{\hat{n}}{2S}}\,\hat{a},\ \hat{a}^+\sqrt{1-\frac{\hat{n}}{2S}}\right] =$$

$$=(2S)\left(\sqrt{1-\frac{\hat{n}}{2S}}\,\underbrace{\hat{a}\hat{a}^+}_{1+\hat{a}^+\hat{a}=1+\hat{n}}\sqrt{1-\frac{\hat{n}}{2S}} - \hat{a}^+\sqrt{1-\frac{\hat{n}}{2S}}\sqrt{1-\frac{\hat{n}}{2S}}\,a\right) =$$

$$=(2S)\left(\sqrt{1-\frac{\hat{n}}{2S}}\,(1+\hat{n})\sqrt{1-\frac{\hat{n}}{2S}} - \hat{a}^+\left(1-\frac{\hat{n}}{2S}\right)\hat{a}\right) =$$

but $\hat{n}\hat{a} = \underbrace{\hat{a}^+\hat{a}}_{\hat{a}\hat{a}^+-1}\hat{a} = (\hat{a}\hat{a}^+-1)\hat{a} = \hat{a}\hat{a}^+\hat{a} - \hat{a} =$
$= \hat{a}\hat{n} - \hat{a}$

$$=(2S)\left((1+\hat{n})\left(1-\frac{\hat{n}}{2S}\right) - \left[\hat{a}^+\hat{a} - \frac{\hat{a}^+}{2S}(\hat{a}\hat{n}-\hat{a})\right]\right) =$$

$$=(2S)\left(\underbrace{(1+\hat{n})\left(1-\frac{\hat{n}}{2S}\right)}_{1+\hat{n}-\frac{\hat{n}}{2S}-\frac{\hat{n}\hat{n}}{2S}} - \left[\hat{n} - \frac{1}{2S}(\hat{n}\hat{n}-\hat{n})\right]\right) =$$

$$=(2S)\left(1+\cancel{\hat{n}} - \frac{\hat{n}}{2S} - \cancel{\frac{\hat{n}\hat{n}}{2S}} - \left(\cancel{\hat{n}} - \cancel{\frac{\hat{n}\hat{n}}{2S}} + \frac{\hat{n}}{2S}\right)\right) =$$

$$=(2S)\left(1-\frac{\hat{n}}{S}\right) = 2(S-\hat{n}) = 2S^z \text{ indeed!}$$

Check $[S^+, S^z] = [S^x + iS^y, S^z] = i\,\varepsilon^{xzy} S^y + i^2 \varepsilon^{yzx} S^x$

$= -S^x - iS^y = -S^+$

$\Rightarrow [S^z, S^+] = S^+$

On the other hand: $[S^z, S^+] = [S - \hat{n}, \sqrt{2S}\sqrt{1-\frac{\hat{n}}{2S}}\,\hat{a}] =$

$= -\sqrt{2S}\left(\hat{n}\sqrt{1-\frac{\hat{n}}{2S}}\,\hat{a} - \sqrt{1-\frac{\hat{n}}{2S}}\,\hat{a}\,\hat{n}\right) = \left(\hat{a}\hat{n} = \hat{n}\hat{a} + \hat{a}\right)$

$= -\sqrt{2S}\left(\sqrt{1-\frac{\hat{n}}{2S}}\,\hat{n}\,\hat{a} - \sqrt{1-\frac{\hat{n}}{2S}}\left(\hat{n}\hat{a} + \hat{a}\right)\right) = \sqrt{2S}\sqrt{1-\frac{\hat{n}}{2S}}\,\hat{a}.$

Ok

The remaining $[\hat{S}^z, \hat{S}^-] = -\hat{S}^-$

can be checked as well, or obtained from one above by hermitian conjugation.

P4 Spin waves in antiferromagnet have <u>linear</u> dispersion

$$\varepsilon_k = \hbar\omega_k = \hbar v k$$

at small momenta $\vec{k}$ (measured from antiferromagnetic $\vec{Q} = \frac{\pi}{a}(1,1,1)$).

There two spin waves — α_k and β_k — which are two polarizations.

Thus $E = 2\int \frac{d\vec{k}}{(2\pi)^3}\, \varepsilon_k\, n(\varepsilon_k) = 2\int \frac{d\vec{k}}{(2\pi)^2} \frac{\hbar v k}{e^{\frac{\hbar v k}{T}}-1}$

let $\frac{\hbar v k}{T} = x \Rightarrow E = 2\frac{(4\pi)}{(2\pi)^3}\int dk\, k^2 \frac{Tx}{e^x-1}$

$k = \frac{T}{\hbar v} x$

$$= \frac{T^4}{(\hbar v)^3}\frac{1}{\pi^2}\int_0^\infty dx \frac{x^3}{e^x-1}$$

$\Rightarrow C_v \sim T^3$.

Magnetization $\langle S_r^z\rangle = S - \langle a_r^+ a_r\rangle = S - \frac{1}{N}\sum_{k_1,k_2} e^{ir(k_1-k_2)}\langle a_{k_1}^+ a_{k_2}\rangle$

But $\langle a_{k_1}^+ a_{k_2}\rangle = \left\langle (ch\theta_{k_1}\alpha_{k_1}^+ - sh\theta_{k_1}\beta_{-k_1})(ch\theta_{k_2}\alpha_{k_2} - sh\theta_{k_2}\beta_{-k_2}^+)\right\rangle$

$= (ch^2\theta_k + sh^2\theta_k)\langle\alpha_k^+\alpha_k\rangle + \underbrace{sh^2\theta_k}_{\text{this is T-independent correction}}$

using $\langle\beta_k^+\beta_k\rangle = \langle\alpha_k^+\alpha_k\rangle = \frac{1}{e^{\frac{\varepsilon_k}{T}}-1}$

But $ch^2\theta_k + sh^2\theta_k = ch 2\theta_k = \frac{A_k}{\sqrt{A_k^2 - B_k^2}} \cong \frac{J}{vk}$ for $k\to 0$.

$\Rightarrow \langle S_r^z\rangle \approx S - \int \frac{d^3\vec{k}}{(2\pi)^2}\frac{J}{vk}\frac{1}{e^{\frac{\hbar v k}{T}}-1} = S - (\text{const})\, T^2$!